ARISTA Access Node KR-ACC-56 Routing Protocol Flap SOP

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5669

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the ARISTA Access Node KR-ACC-56. The node, located at KR Puram Access, is a critical component of the network infrastructure, and its failure can have far-reaching consequences. The protocol flap failure domain occurs due to the complex interplay between the BGP, OSPF, and ISIS routing protocols, which are used to manage and optimize network traffic flow. In the context of the ARISTA Access Node, the routing protocol flap can be caused by a variety of factors, including misconfigured BGP peers, OSPF adjacency issues, or ISIS LSP flooding. The failure of the routing protocol flap can result in network instability, packet loss, and increased latency, ultimately impacting the quality of service experienced by users. To mitigate this risk, it is essential to identify and address the root causes of the routing protocol flap failure as quickly and efficiently as possible.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and power is available to the node.

- Confirm that the redundant link status is operational and can take over in case of a failure.

- Ensure that remote hands are dispatched to the site to assist with the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
Misconfigured BGP peers	High
OSPF adjacency issues	Medium
ISIS LSP flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 0, main routing table version is 0, Description: This command displays the BGP summary, including the table version and main routing table version. The output will help to identify any issues with the BGP peers or routing table.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor status, including the neighbor ID, priority, state, dead time, address, and interface. The output will help to identify any
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issues with the OSPF adjacency.

Command: show isis database, Expected Output: ISIS Level 2 Database for node KR-ACC-56, Description: This command displays the ISIS database, including the level 2 database for the node. The output will help to identify any issues with the ISIS LSP flooding.
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Resolution Steps

Step 1: Step: Verify BGP peer configuration, Command: show bgp peers, Description: Verify the BGP peer configuration and ensure that all peers are correctly configured. If any issues are found, correct the configuration and save the changes.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf neighbor , Description: Reset the OSPF adjacency by clearing the neighbor cache and re-establishing the adjacency. This may resolve any issues with the OSPF adjacency.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp-flood , Description: Clear the ISIS LSP flooding by clearing the LSP flood cache and re-establishing the LSP. This may resolve any issues with the ISIS LSP flooding.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version is 0, main routing table version is 0, Description: Verify that the BGP summary is correct and there are no issues with the BGP peers or routing table.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify that the OSPF neighbor status is correct and there are no issues with the OSPF adjacency.

Command: show isis database, Expected Output: ISIS Level 2 Database for node KR-ACC-56, Description: Verify that the ISIS database is correct and there are no issues with the ISIS LSP flooding.
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Roll-back Procedures

- Step: Restore previous BGP peer configuration, Command: load merge /config/bgp/previous-configuration, Description: Restore the previous BGP peer configuration from the backup file.

- Step: Restore previous OSPF adjacency configuration, Command: load merge /config/ospf/previous-configuration, Description: Restore the previous OSPF adjacency configuration from the backup file.

- Step: Restore previous ISIS LSP flooding configuration, Command: load merge /config/isis/previous-configuration, Description: Restore the previous ISIS LSP flooding configuration from the backup file.